

v1.0OEU_ Benevolent Particle Overpressure

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: OEU Topological Engineer’s Handbook **Type:** Scientific Change Log | **Date:** June 2026
Applies to: v1.0OEU_ Benevolent Particle Overpressure V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{17} + particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Schwinger pair production	E_crit stated only	LUXE 2024: first experimental approach to E_Schwinger/100. Strong-field QED now testable	HIGH	Schwinger physics transitioning from theoretical to experimental — update probabilities
Vacuum stability	V1.0 era landscape	Di Luzio 2023: lifetime > 10^{600} yr. V2.0: VACUUM_BUBBLE event probability ~ 0 (current epoch)	MEDIUM	Vacuum stability parameter updated; event probability now quantified
Unruh radiation	Mentioned theoretically	2023 analog system: first evidence for Unruh thermal distribution. V2.0: UNRUH_PHOTON events for high-acceleration particles	LOW	Lab confirmation adds physical grounding to Unruh radiation in simulation

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Hawking point	Not covered	Penrose 2024: 5 CCC Hawking radiation circles in Planck CMB. V2.0: CCC_SIGNAL background field added	LOW	Cyclic cosmology observable suggested — adds background field type

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Schwinger
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0OEU_Benevolent Particle Overpressure V1.0 archived | V2.0 is the active specification as of June 2026